Widgets, their types and parameters

Project: `code/week04/examples`, files 01 to 08

Configuring a widget

You configure a widget with the parameters of its constructor, almost all **named**: `name: value`.

```
Text('Hello',                  // positional
  textAlign: TextAlign.center,  // named
  style: TextStyle(fontWeight: FontWeight.bold))
```

Simple value	`width: 120`, a string, a `bool`
Enum constant	`TextAlign.center`, `BoxFit.contain`
Object or widget	`child: Text(..)`, `EdgeInsets.all(8)`
Function	`onPressed: () { ... }`

Parameter	Holds
`child`	*one* widget: `Center`, `Padding`, `Container`, `Card`, buttons
`children`	a *list*: `Row`, `Column`, `Stack`

Find parameters: `Ctrl+Space` inside the parentheses; `Ctrl+click` the name; `api.flutter.dev`. `required` ones must be given.

Text and TextStyle

`style`	a `TextStyle`
`textAlign`	`left right center justify start end`
`maxLines`	maximum number of lines
`overflow`	`clip ellipsis fade visible`
`softWrap`	`false`: one line only

`TextStyle`	Example
`fontSize`	`28`
`fontWeight`	`FontWeight.bold` (or `.w100` to `.w900`)
`fontStyle`	`FontStyle.italic`
`color`	`Colors.teal`
`letterSpacing`	`1.5`
`decoration`	`TextDecoration.underline`

`textAlign` **aligns lines inside the *width of the Text*. In a** `Column` **with** `crossAxisAlignment: start` **the text is only as wide as its letters: nothing moves. Use** `SizedBox(width: double.infinity)` **or** `Center`**.**

Buttons

`ElevatedButton`	filled, shadow: the main action
`TextButton`	text only: a discreet action
`OutlinedButton`	thin frame: an alternative

Parameters: `child` (label), `onPressed`, `onLongPress`, `style`. `onPressed: null` ⇒ the button is **disabled** (grey).

```
ElevatedButton(
  onPressed: () => print('tapped'),
  style: ElevatedButton.styleFrom(
    backgroundColor: Colors.deepPurple,
    foregroundColor: Colors.white,
    elevation: 8),
  child: const Text('Elevated'),
)
```

Obsolete: `RaisedButton`, `FlatButton`, `OutlineButton`, `primary`, `onPrimary`.

Image

```
Image.asset('assets/logo.png') // project file
Image.network('https://...')   // a URL
Image.file(file)               // on the device
Image.memory(bytes)            // in memory
```

`width`, `height` (only one: proportions kept), `fit`, `color` (tint), `alignment` (what stays when cropped).

`BoxFit`	Effect
`contain`	all visible, proportions kept, gaps possible
`cover`	box filled, proportions kept, may crop
`fill`	box filled, image stretched
`fitWidth/Height`	width (height) matches the box
`scaleDown`	like `contain`, never enlarged (default)

Assets: declare the folder in `pubspec.yaml`, then `flutter pub get` or restart.

```
flutter:
  assets:
    - assets/
```

Icon

```
Icon(Icons.favorite, color: Colors.red, size: 40)
```

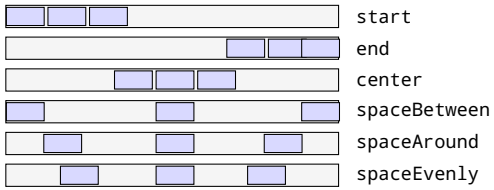
First parameter: an `IconData` (`Icons.home`, ...); `size` is 24 by default.

Copyrighted content Dr Mohamad AOUDE

Row and Column

Main axis: the direction of the widget (horizontal for Row). **Cross axis**: the other one.

`mainAxisAlignment`	free space *along* the main axis
`crossAxisAlignment`	`start`, `center` (default), end, `stretch`
`mainAxisSize`	`max` (default) or `min`



`spaceAround`: gaps between children are twice the end gaps; `spaceEvenly`: all gaps equal. A `Column` needs extra height to have free space.

Padding, Stack and helpers

`Padding(padding: ..., child: ...);` `padding` is required:

```
EdgeInsets.all(8)          // all four sides
EdgeInsets.symmetric(
  horizontal: 16, vertical: 8)
EdgeInsets.only(left: 4, top: 12)
EdgeInsets.fromLTRB(1, 2, 3, 4) // L T R B
```

`Stack(alignment: Alignment.center, children: [...])`: last child on top. Exact place: `Positioned(top: 8, right: 8, child: ...)`.

`SizedBox`	fixed size; empty space
`Card`	`elevation`, `color`, `margin`
`SingleChildScrollView`	makes a long page scroll

Scaffold slots

`appBar`	bar at the top
`body`	content of the page
`floatingActionButton`	round floating button
`bottomNavigationBar`	bar at the bottom
`drawer` / `endDrawer`	side menu, left / right
`backgroundColor`	color of the page

- Fills the screen; the body shrinks by itself when the keyboard opens (`resizeToAvoidBottomInset: true`).
- Notched bar: `BottomAppBar(shape: CircularNotchedRectangle())` and `FloatingActionButtonLocation.centerDocked`. Keep text away from the middle.
- A button that changes the page needs a stateful page and `setState`.

Material and Cupertino

Material	Cupertino
`MaterialApp`	`CupertinoApp`
`Scaffold`	`CupertinoPageScaffold`
`AppBar (title)`	`CupertinoNavigationBar (middle)`
`body`	`child`
`ElevatedButton`	`CupertinoButton`
`BottomNavigationBar`	`CupertinoTabScaffold` + `CupertinoTabBar`
`Drawer`	none (no drawer on iOS)

There is no `CupertinoScaffold`. Imports: `material.dart`, `cupertino.dart`.

A drawer

```
drawer: Drawer(
  child: ListView(
    padding: EdgeInsets.zero, // header at top
    children: [
      const DrawerHeader(child: Text('Menu')),
      ListTile(title: const Text('Settings'),
        onTap: () => Navigator.pop(context)),
    ]),
),
```

- The `AppBar` adds the menu button by itself; a swipe also opens it.
- It does **not** close on a tap: call `Navigator.pop(context)`.
- `ListTile`: `leading`, `title`, `subtitle`, `trailing`, `selected`, `onTap`.
- Own button: `Scaffold.of(context).openDrawer()` with a context *below* the `Scaffold` (wrap in a `Builder`).

An AppBar in its own class

```
class MyAppBar extends StatelessWidget
    implements PreferredSizeWidget {
  const MyAppBar({super.key,
                  required this.title});
  final String title;
  @override
  Size get preferredSize =>
      const Size.fromHeight(100);
  @override
  Widget build(BuildContext context) =>
      AppBar(title: Text(title),
             toolbarHeight: 100);
}
```

The `appBar` slot needs the height first: give the **same** number to `preferredSize` and `toolbarHeight`. Dark background ⇒ set `foregroundColor`.
Since Dart 2.17, `const MyApp({super.key});` is the short form of `const MyApp({Key? key}) : super(key: key);`.

Copyrighted content Dr Mohamad AOUDE

Expanded with a ListView

```
Column(children: [
  header,
  Expanded(child: ListView.builder(
    itemCount: 50,
    itemBuilder: (context, i) =>
        ListTile(title: Text('Item $i')))),
  footer,
])
```

- Without `flex`, `Expanded` children share the space **equally**.
- A `ListView` wants all the height; in a `Column` that is unlimited ⇒ error. `Expanded` gives it exactly the rest.
- `ListView.builder` builds rows only when they appear: long lists.
- `RichText` + `TextSpan`: parts with different styles.

Container and BoxDecoration

`margin`	space *outside* the rectangle
`padding`	space *inside*, around the child
`alignment`	where the child sits
`transform`	`Matrix4.rotationZ(-0.1)` (radians)
`decoration`	a `BoxDecoration`

```
Container(
  margin: const EdgeInsets.all(16),
  padding: const EdgeInsets.all(16),
  decoration: BoxDecoration(
    color: Colors.white,
    border: Border.all(color: Colors.blue,
                       width: 3),
    borderRadius: BorderRadius.circular(16),
    boxShadow: [BoxShadow(blurRadius: 8,
      color: Colors.black38,
      offset: Offset(4, 4))],
    // gradient: LinearGradient(colors: [..])
    // shape: BoxShape.circle
  ),
  child: const Text('A card'),
)
```

- Never `color` *and* `decoration`: put the color in the decoration.
- Circle: equal `width` and `height`, `BoxShape.circle`, no `borderRadius`.
- No child and no size: takes all the space; with a child: its size.

Common errors

Vertical viewport was given unbounded height	`ListView` in a `Column`: wrap it in `Expanded`
Invisible title in the bar	set `foregroundColor`
Own bar refused by `appBar`	implement `PreferredSizeWidget`
Button does not change the screen	stateful page, `setState`
Text under the docked button	move it left or right
Drawer stays open	`Navigator.pop(context)` in `onTap`
Scaffold.of() called with a context that does not contain a Scaffold	use a `Builder`
Cannot provide both a color and a decoration	put the color in the `BoxDecoration`

The programs

```
flutter run -t lib/01_widgets_full_program.dart
```

01	full program: all widgets and parameters; remembers the last button
02	`Scaffold`: notch, docked button, keyboard
03	custom `AppBar`: `PreferredSizeWidget`
04	four equal `Expanded`; `RichText`
05	`ListView` between header and footer
06	`drawer`, `endDrawer`
07	the Cupertino widgets
08	`Container`: `BoxDecoration`, circle, `transform`

Each file starts with a comment: what you learn and what you should see.

Copyrighted content Dr Mohamad AOUDE